

CSE 4849 / CSE 4885
Human Computer Interaction (HCI)
Lecture 0: Course Introduction

Summer 2025

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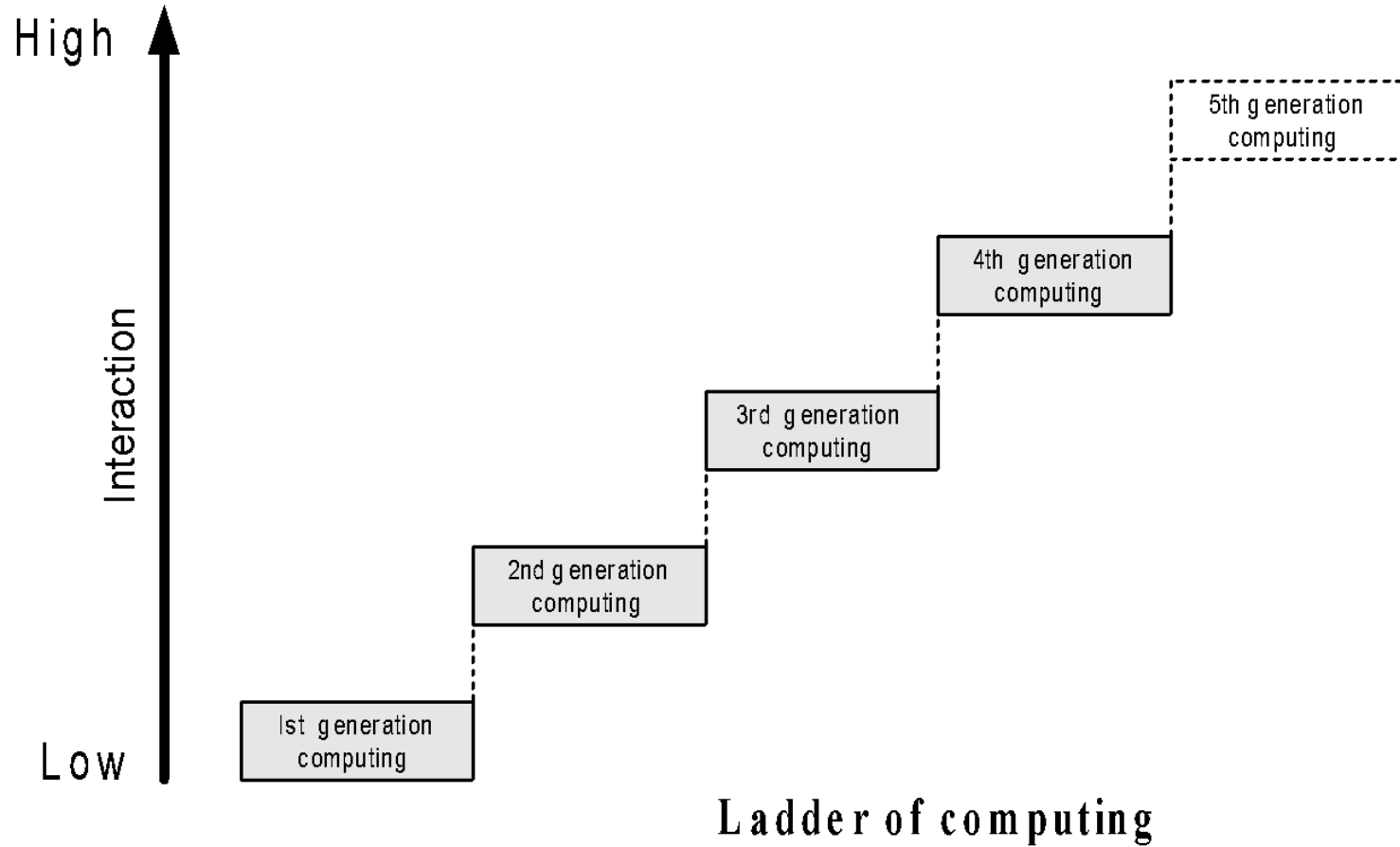
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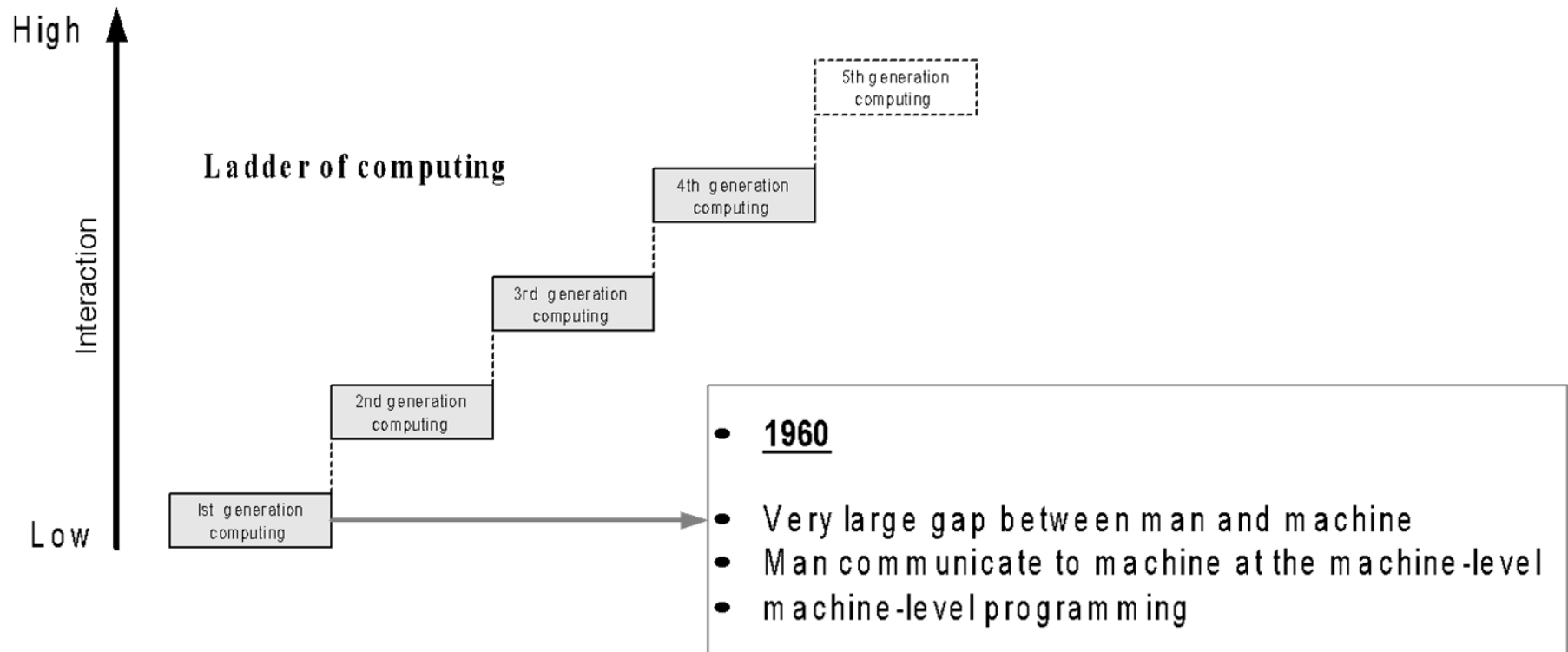
Genesis

- **Shift of focus from**
 - System-centered computing
 - Almost no interaction
 - Computer = hardware + machine-level code
- to
- Human-centered computing
 - Very high-level interaction
 - Computer = hardware + software + algorithm

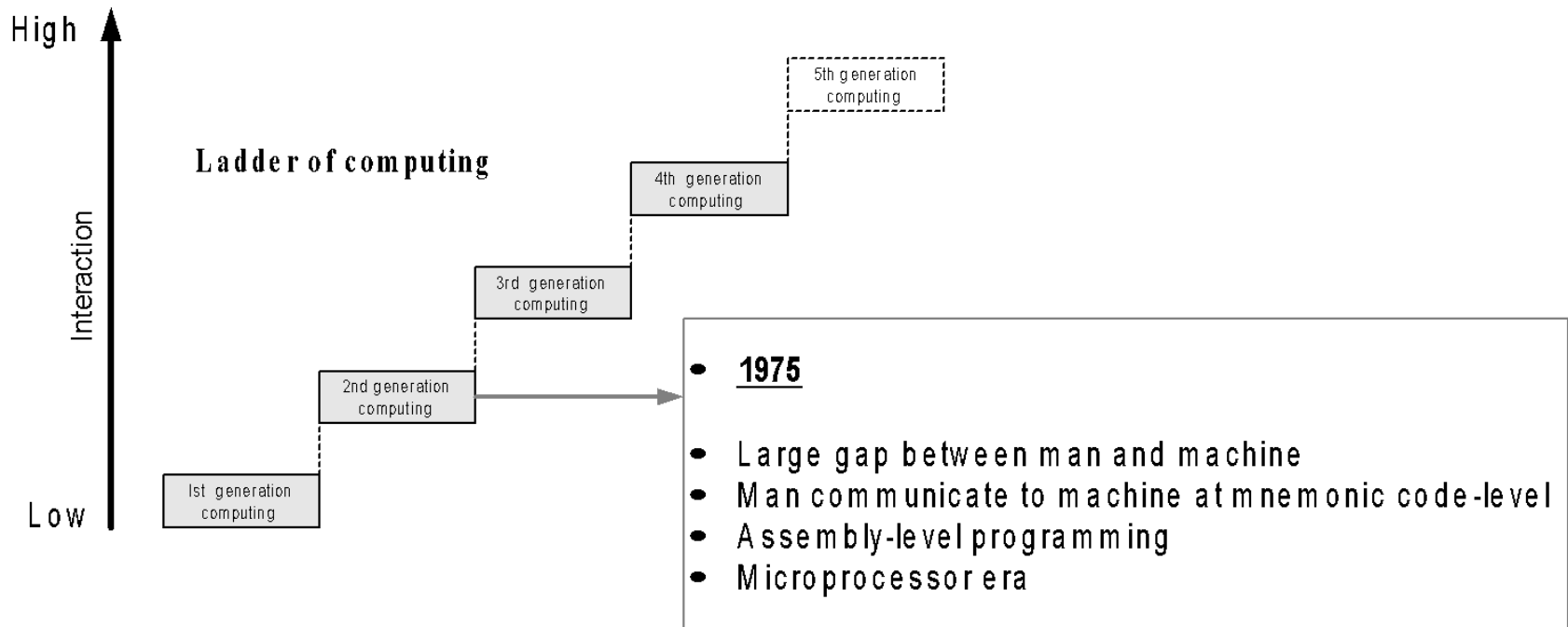
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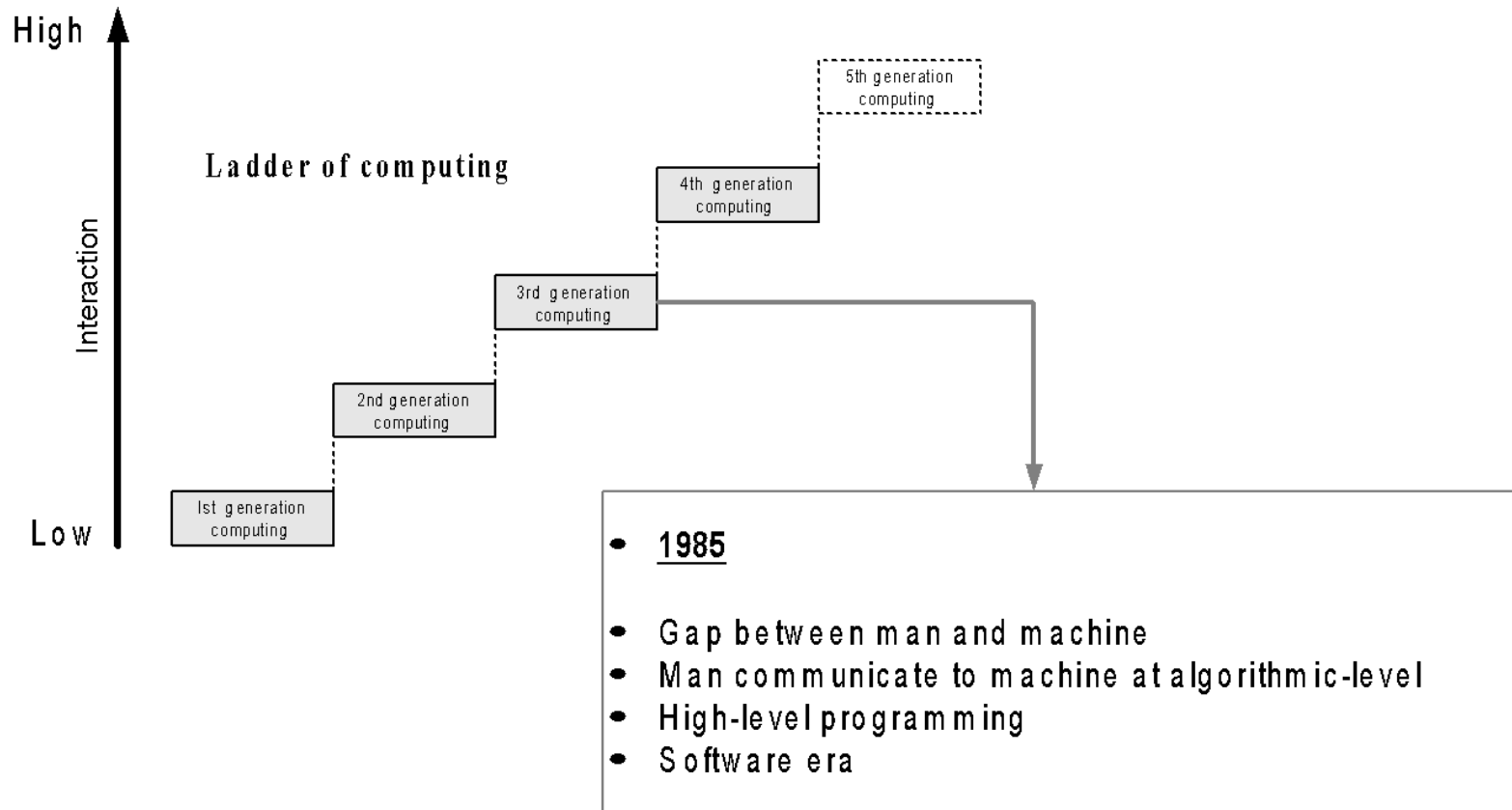
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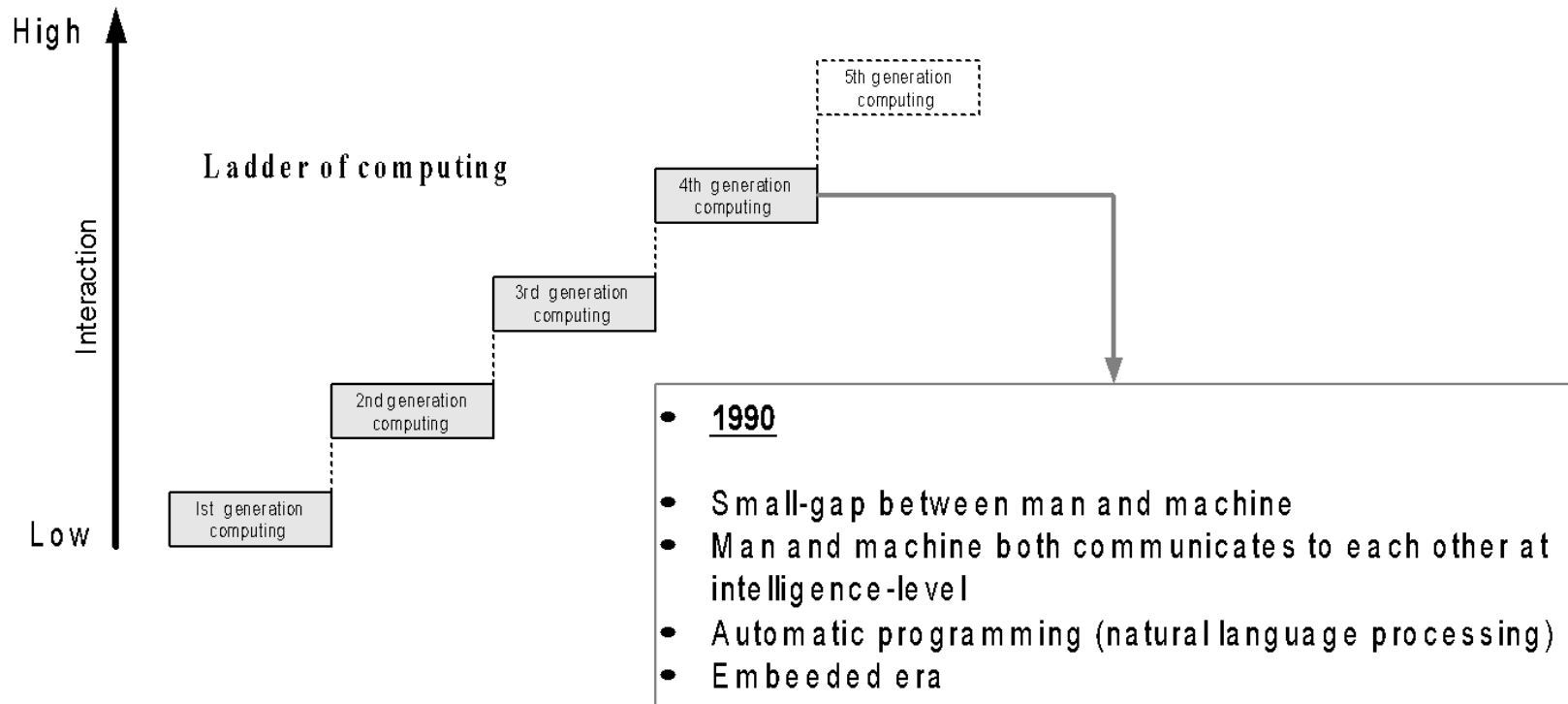
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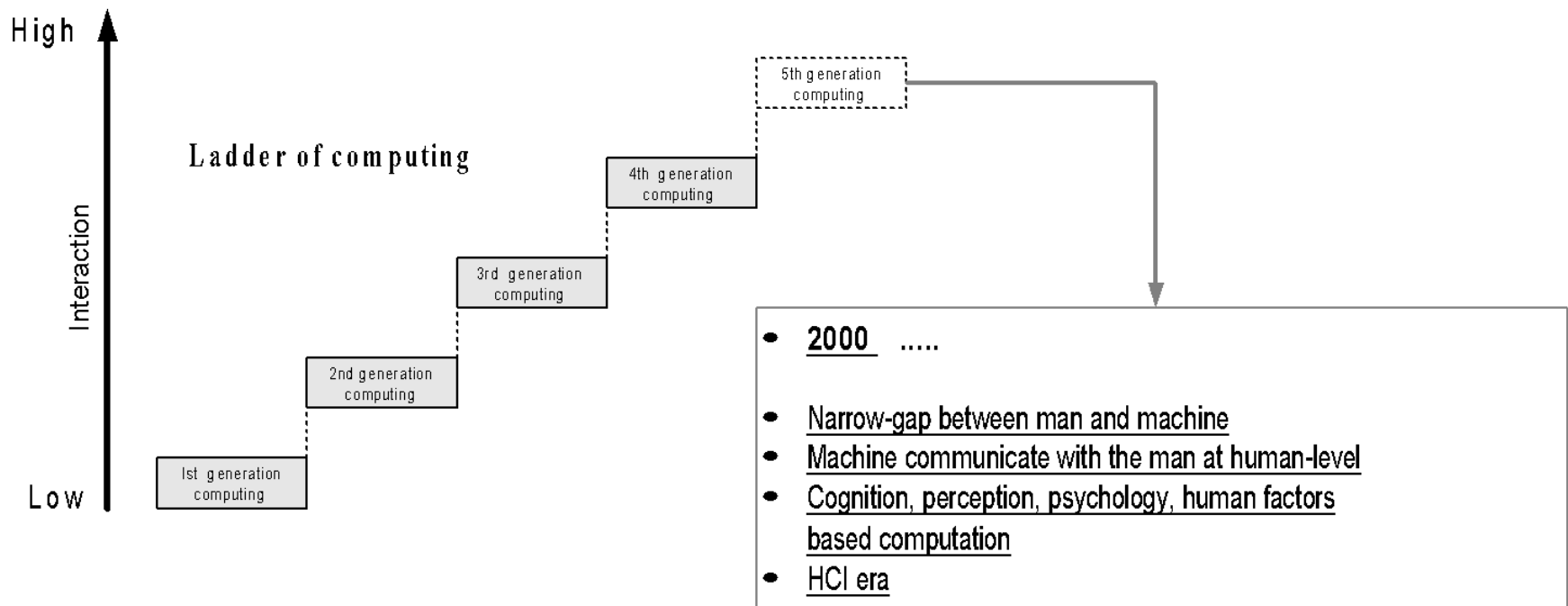
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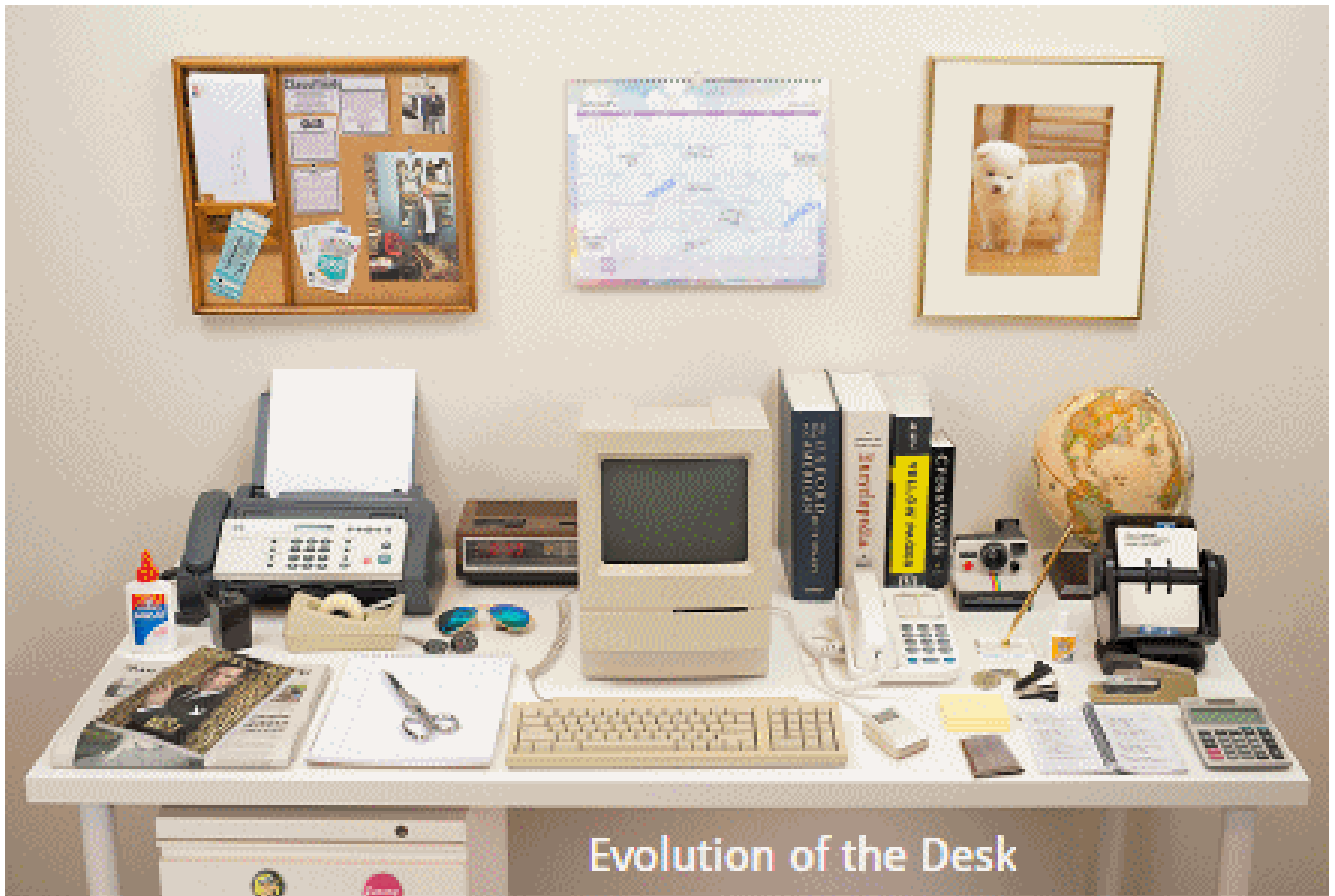


Genesis



Genesis



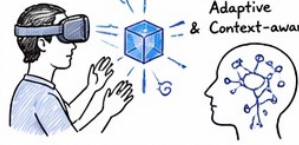


Evolution of the Desk

1980

EVOLUTION OF HUMAN-COMPUTER INTERACTION (HCI)

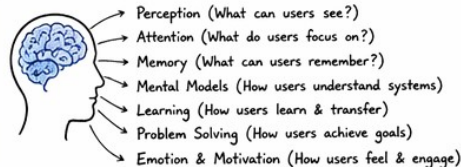
From Reducing the Gap to Creating Seamless, Intelligent, and Human-Centered Experiences

TIME	Pre-1980s (Pre-Computer)	1980s (Emergence)	1990s (Expansion)	2000s (Mobility & Web)	2010s (Natural & Contextual)	2020s+ (Intelligent & Immersive)
KEY INTERFACE EXAMPLES	Punch Cards, Paper Forms, Teletypes 	Command-Line Interfaces (CLI) Early GUIs (Macintosh, Windows) 	Graphical Interfaces Mature, WIMP Web Browsers (WWW) 	Mobile Devices & Applications Web 2.0, Rich Internet Apps 	Touch Interfaces, Smartphones, Tablets Voice Assistants, Wearables 	XR Interfaces (AR/VR/MR) AI-driven, Adaptive & Context-aware 
INTERACTION PARADIGM SHIFT	No Direct Interaction	Syntax-Based Interaction	Direct Manipulation (Visual)	Networked Interaction (Information Access)	Mobile, Multi-touch, Multi-modal	Immersive, Conversational, Ambient Intelligence
FOCUS OF HCI	Automation of Computation	Efficiency, Accuracy, Syntax & Commands	Usability, Learnability, Consistency	Accessibility, Connectivity, Portability	Natural Interaction, User Experience (UX), Context-Awareness	Human-AI Collaboration, Personalization, Presence & Trust
COMPUTER BECOMES ...	Centralized (Mainframes)	Personal (Desktops)	Connected (Networks)	Mobile (Cloud)	Ubiquitous (Everywhere)	Intelligent (Embedded AI)
INTERACTION GAP (Man-Machine)	Very Large Gap	Large Gap	Medium Gap	Small Gap	Very Small Gap	Seamless / Symbiotic

HCI GUIDING PRINCIPLES (Across All Eras)

- Visibility & Feedback (Norman)
- Affordances & Constraints
- Consistency & Standards
- User Control & Freedom
- Error Prevention & Recovery
- Match Between System & Real World
- Recognition Rather Than Recall
- Flexibility & Efficiency of Use
- Aesthetic & Minimalist Design
- Help & Documentation

COGNITIVE & PSYCHOLOGICAL FOUNDATIONS IN HCI



Goal: Reduce Cognitive Load, Increase Effectiveness & Satisfaction

KEY DRIVERS OF EVOLUTION

- Advances in Technology (Processing, Sensors, AI, Networks)
- Changes in Human Needs & Work Practices
- Theories from Cognitive Science, Psychology, Sociology
- Design & Evaluation Methods (User-Centered, Iterative)
- Societal & Cultural Context

THE NORTH STAR

Create technologies that empower people by aligning with how they think, feel, and behave.

HCI is not just about technology – it is about understanding humans and designing meaningful interactions.

Towards the reality

- How more accurately and efficiently the behavior of human can be simulated in a computer?

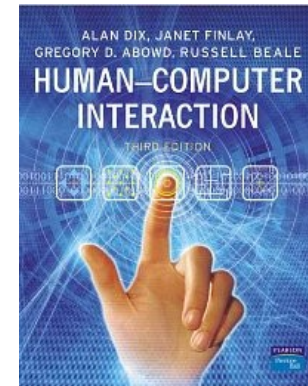
Introduction

- Why interested in HCI class?
- What is your strength?
 - Designer?
 - Sophisticated Programmer?
 - Good psychologist?

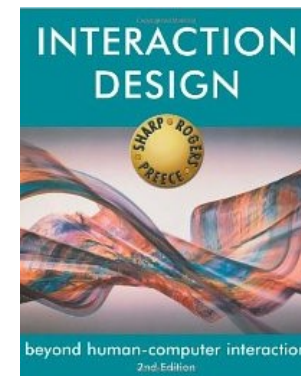
Course Information

- Books

- *Human-Computer Interaction*, by Alan Dix, Janet Finlay, Gregory Abowd, and Russell Beale. Prentice Hall, 2004.



- *Interaction Design-beyond Human Computer Interaction*, by Yvonne Rogers, Helen Sharp, Jenny Preece, Wiley, 6th edition, 2016.



Grading

▪ Class attendance	: 30
▪ Assignments*	: 45
▪ Mid-semester Exam	: 75
▪ Semester-final Exam	: 150
▪ Total	: 300

*explained in the next slide

Assignments

- 4 assignments will be given
- Assignments will be counted as quizzes
- **Topic Selection**
 - Research paper readings on HCI for Project topic selection
 - Identify team and general topic
 - Suggestion: Pick a population and pick a technology; check out intersection

Assignments...

- **Assignment 1: Need finding**
 - Identifying the user needs and establishing requirements
 - Understanding the problem space
 - Describe tasks, users, environment, social context
 - What are implications for design?
- **Assignment 2: Develop design alternatives**
 - Design Alternatives
 - Storyboard, scenarios, mockups for alternative designs (multiple different designs)
 - Explore, push boundaries of design space
 - Explain decisions rationally

Assignments...

- **Assignment 3: Prototype implementation**
 - System prototype and evaluation plan
 - More detail Implementation of the prototype
 - Plan for conducting real evaluation
- **Assignment 4: Evaluation**
 - Evaluation
 - Conduct formal evaluation with example users
 - Use appropriate methods
 - Analyze results of evaluation
 - Characterize what's working and what's not

Presentation and Report

- For each of the assignments students need to present their part through out the class period
 - Idea discussion
 - Feedback
 - Contribution
- Each team should submit a report formatted in LaTeX (for good mark) within submission deadline.
- Failure to submit report or presentation will cause for grade reduction.

Project teams

- 3-5 people
 - You decide (or I will!)
 - Diverse is best!
 - Use the personal blog (or as you like any)
 - Immediately post ideas for general topics, progress, technologies, problems, solutions etc.
- Cool Project and team name

Objectives of the course

(loosely based on ACM-SIGCHI HCI Curricula)

- Gain understanding of
 - The scope and the issues affecting HCI
 - The importance of the user interface and the impact of good and bad user interfaces
 - The limits of the individuals developing HCI systems and the need to work in team
 - How HCI concerns can be incorporated into the systems development cycle
 - The research branches of HCI

Objectives of the course...

- Have some knowledge of
 - Methodologies for HCI design
 - Namely, User-Centred Design (Interaction Design)
 - Techniques for HCI design
 - Contextual Inquiry, Affinity Diagrams, Ethnography, ...
 - Techniques for evaluation of interfaces
 - Expert evaluation, experimental design

HCI Research



IBM Research

- Human Computer Interaction Research at IBM has a vibrant global community, and a long history. Some of their larger research groups and centers are:
 - Visualization
 - Cognitive & user modeling
 - Usable mobile security
 - Mobile collaboration and learning
 - Smarter cities
- IBM has completely moved to HCAI (<https://research.ibm.com/topics/human-centered-ai>)

HCI Research...

Microsoft Research

- Human-Computer Interaction (@ MSRA)
 - Microsoft Research's Human-Computer Interaction Group (HCI@MSR) comprises a world-renowned, interdisciplinary team of research scientists, engineers, and designers who take a user-centered approach to developing, designing, and studying computing technology and its use.
- Some projects:
 - Teachable AI Experiences (Tai X)
 - Project Zanzibar
- <https://www.microsoft.com/en-us/research/group/human-computer-interaction/projects/>

HCI Research...



- HCI research has fundamentally impacted the design of Gmail and Android, and Google are engaged in a variety of HCI areas such as
 - Mobile and ubiquitous computing,
 - Social computing,
 - Interactive visualization and visual analytics, as well as search and browsing interfaces.
- HCI Projects include
 - Gesture and touch interaction;
 - Activity-based and context-aware computing;
 - Recommendation of social and activity streams;
 - Analytics of social media engagements, and
 - End-user programming.
- As Google at large we follow a simple but vital premise: “Focus on the user and all else will follow.”
- <http://research.google.com/about.html>
- <https://research.google/research-areas/human-computer-interaction-and-visualization/>

HCI Research...

- Meta/Facebook: AR/VR-based interactions
- Apple: Gesture-based interaction in mix reality environment
- Netflix: Interactive content analysis
- Disney: digital storytelling and user engagement
- Nvidia: Gaze estimation, graphics and gaming
- Tesla: Autopilot interface, in-car touchscreen UI
- SSL: Head mounted mouse controller, ASD recognition, HCAI/HCNLP, Bangla word-level/sentence-level sign recognition/translation, mental health status recognition/conversational AI, RLHF, HITL, etc.

STANFORD HCI GROUP

<http://hci.stanford.edu/>

B*i*D

Berkeley
institute
of design

<http://bid.berkeley.edu/>



<http://www.cs.umd.edu/hcil/>

Carnegie Mellon



• Human-
Computer
Interaction
Institute

<http://www.hcii.cmu.edu/>

<http://hci.cornell.edu/>



Cornell University

SEARCH CORNELL

HCI, CMC, and CSCW Design at Cornell University

For knowing Country wise HCI research :
<http://degraaff.org/hci/groups.html>

HCI conferences and journals

- Top Conferences:
 - CHI - Computer Human Interaction, ACM SIGCHI
 - CSCW - Conference on Computer Supported Cooperative Work
- Top Journals:
 - IJMMS - International Journal of Human-computer Studies / International Journal of Man-machine Studies [Elsevier]
 - *Computers in Human Behavior* [Elsevier]
 - TOCHI - ACM Transactions on Computer-human Interaction
 - IEEE Transaction on Man-Machine Systems
 - IJHCI, IJHCS